

From CT-CMT to XVA — Multi-Task Continual TTA as a Structured Extension of Our Prior Work

1. Motivation and continuity with the initial work

Our prior work (Moraiti *et al.*, *EJAI* 2026 — “Continual Test-Time Domain Adaptation for Object Detection via Contrastive Mean Teacher and Stochastic Restoration”) established a three-component recipe for source-free continual TTA on a single detection task:

1. **Mean Teacher (MT)** with an EMA of the student weights, $\theta_T \leftarrow \alpha\theta_T + (1 - \alpha)\theta_S$;
2. **Object-level Contrastive Learning (CL)** — SupCon (Khosla *et al.* 2020) over per-object RoI-aligned features $z_T(b)$, $z_S(b)$ pulled by the teacher’s own pseudo-labels;
3. **Stochastic Restoration (SR)** — Bernoulli(p) restoration of a random fraction of student weights to source (CoTTA convention, uniform mask).

The evaluation on SHIFT, KITTI, Cityscapes, CLAD-D, and COCO-C showed that **SR was the dominant contributor to preserving source knowledge** across long continual streams and that adding object-level CL on top of MT was a consistently positive but smaller boost. Two failure modes were left open by that formulation:

- Object-level CL uses only *one head’s* view per object (RoI features). When the teacher becomes wrong about the class, the SupCon loss silently reinforces the mistake — there is no orthogonal signal to catch it.
- SR restores every parameter with the same Bernoulli probability. The backbone (task-invariant, gradient-heavy, shared across every loss term) is thus under-corrected relative to task-specific layers.

This paper extends the CT-CMT framework to multi-task learning and, in doing so, closes both failure modes at the mechanistic level. The extension generalises the two components that carry over (CL $\rightarrow$ CT-CL, SR $\rightarrow$ STR), adds two mechanisms that only exist in the MTL setting (CT-CR, CTPV), and inherits the MT skeleton unchanged.

Everything in the paper is either (i) a re-implementation of a component already in Moraiti *et al.* 2026 for a new backbone, (ii) a strict generalisation that reduces to the prior work when the MTL setting is collapsed, or (iii) a genuinely new mechanism that is only *possible* in MTL. §10 summarises the mapping.

2. Prior recipe, formalised

Let b be a pseudo-detected bounding box of class c produced by the teacher’s own detection head. In Moraiti *et al.* 2026 the CL loss reads

$$\mathcal{L}_{\text{CL}} = \frac{\lambda}{N} \sum_{i=1}^N \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(z_i^S \cdot z_p^T / \tau)}{\sum_{j=1}^N \exp(z_i^S \cdot z_j^T / \tau)}$$

with:

- $z_i^M = \text{Normalize}(\text{RoIAlign}(F^M, b_i))$, where $M \in \{S, T\}$ is student or teacher and F is a multi-scale backbone/PAFPN feature map;
- $P(i) = \{p : C_p = C_i\}$ the same-class positive set;
- $\tau = 0.07$, $\lambda = 0.003$.

The **object-level view** z_i is a per-bbox RoIAlign. There is only one view per object — no orthogonal supervision signal from a second head. The loss cannot fire when only one class is

present in the image (all pairs are positive) and cannot distinguish a genuinely well-adapted representation from one that is class-degenerate.

The SR mechanism is described in §3.2 of Moraiti *et al.* 2026 as

$$M \sim \text{Bernoulli}(p), \quad W_{t+1} = M \cdot W_0 + (1 - M) \cdot W_{t+1}$$

applied element-wise to *every* trainable weight in the student. $p = 0.01$ on KITTI/Cityscapes/CLAD-D; $p = 0.025$ on the SHIFT sequences.

3. Cross-Task Contrastive Learning (CT-CL) — a strict generalisation of prior object-level CL

The MTL setting exposes each object at the level of the shared backbone via **two independent heads** — detection and segmentation. Our first extension augments the prior object-level view $z^S(b)$ with a second view $z_{\text{seg}}^S(b)$ constructed from the teacher’s per-pixel segmentation posterior:

$$z_{\text{det}}^S(b) = \phi_{\text{det}}(\text{RoIAlign}(F^S, b)) \quad (\text{as in prior work})$$

$$z_{\text{seg}}^S(b) = \phi_{\text{seg}}\left(\frac{\sum_{i \in b} p_{i, \sigma(c)}^T F_i^S}{\sum_{i \in b} p_{i, \sigma(c)}^T + \varepsilon}\right) \quad (**\text{new}**)$$

where $\sigma(c)$ maps the detection class c to the corresponding segmentation class via the Cityscapes taxonomy, and $\phi_{\text{det}}, \phi_{\text{seg}}$ are lightweight L2-normalised 128-d projection heads. The det view is identical to the prior CT-CMT construction. The seg view is the *teacher segmentation posterior used as an attention mask over student features*, which is why it fires only for objects the seg head recognises and does not require any additional annotation.

The SupCon loss then operates over the combined pool $\{(z_i, y_i)\}$ where $z_i \in \{z_{\text{det}}^S, z_{\text{seg}}^S\}$ and y_i is the detection class. Positive pairs come in three qualitatively different flavours:

Pair type	Effect	Available in Moraiti 2026?
$z_{\text{det}} \leftrightarrow z_{\text{det}}$ (same class)	classical object-level SupCon	yes
$z_{\text{seg}} \leftrightarrow z_{\text{seg}}$ (same class)	new — encourages seg-view coherence	no (single-task)
$z_{\text{det}} \leftrightarrow z_{\text{seg}}$ (same class)	new — forces two heads to converge to a shared per-class embedding	no (single-task)

Prior work as a special case. Deleting the segmentation branch removes z_{seg} , dropping rows 2 and 3 and recovering the object-level CL loss of Moraiti *et al.* 2026 verbatim (up to the concrete backbone — YOLOX PAFPN in prior work, Panoptic-FPN R-50 here). CT-CL is thus a strictly larger family; any conclusion drawn in the prior paper about object-level CL still applies to the row-1 subset of CT-CL.

Why this closes the “silent-mistake” issue. When the teacher’s detection head misclassifies an object as class c' but the seg head still labels the pixels as $\sigma(c)$ (the *true* class), $z_{\text{seg}}(b)$ diverges from $z_{\text{det}}(b)$. The cross-pair SupCon term then produces a large gradient that *cannot* be reconciled by moving both projections toward the same wrong-class centroid:

they are seeded from different heads. The loss surface pushes the shared backbone toward a representation on which both heads agree — which is exactly the class that is present.

4. Cross-Task Consistency Regularizer (CT-CR) — a new mechanism

CT-CL is instance-level (one embedding per bbox). To close the same-image loop at the pixel level, we add a dense cross-task consistency regulariser: every pixel inside a pseudo-box b of class c is treated as a segmentation-side pseudo-label of class $\sigma(c)$:

$$\mathcal{L}_{\text{CT-CR}} = \sum_{i \in b} \text{CE}(p_i^{S, \text{seg}}, \sigma(c))$$

with pixels outside every box ignored (`ignore_index=255`). This propagates detection-side certainty back through the segmentation head and, symmetrically, ensures the segmentation head’s per-pixel predictions be spatially consistent with the detection geometry.

There is **no analog of CT-CR in Moraiti *et al.* 2026** because the prior work has no segmentation head to receive the supervision. It is a novel mechanism only *definable* in the MTL setting. Empirically CT-CR contributes part of the +1.44 mIoU that CT-CL + CT-CR together deliver over the STR-only baseline; it is enabled by default with $\lambda_{\text{CR}} = 0.3$.

5. Shared-Trunk Stochastic Restoration (STR) — a task-aware generalisation of prior SR

Moraiti *et al.* 2026 applies SR uniformly across all trainable weights. In the MTL setting, backbone and PAFPN weights carry gradient traffic from *every* loss term (det + seg + CT-CL + CT-CR), while task-specific head weights receive gradients only from their own task. A uniform Bernoulli mask therefore under-corrects the trunk relative to the heads. STR splits the mask by parameter role:

$$\Pr[\text{restore } \theta_i] = \begin{cases} p \cdot \eta & \text{if } \theta_i \in \text{backbone} \cup \text{PAFPN (shared)} \\ p & \text{otherwise (task-specific heads)} \end{cases}$$

with $p = 0.01$ (matching prior work) and $\eta = 0.1$. The shared trunk gets restored $10\times$ more often than the heads per weight, in expectation.

Prior SR is the $\eta = 1$ special case of STR (uniform mask). When the model is single-task (no distinction between shared and task-specific components), η is meaningless and STR reduces to the prior formulation exactly. The extension is therefore backwards-compatible.

Why this closes the “under-corrected backbone” issue. In the prior paper’s KITTI/COCO-C experiments the “Forgetting” metric tracked the model’s ability to return to source performance after adaptation. With uniform SR, forgetting was small on shorter streams but grew visibly on COCO-C’s 15-domain sequence. STR’s higher backbone-restore rate protects the shared representation against exactly this long-run drift, at zero extra cost to task-specific fine-tuning capacity.

6. Cross-Task Pseudo-Label Verification (CTPV) — a new mechanism

The MT skeleton in Moraiti *et al.* 2026 accepts every teacher-detected box (above a confidence threshold) as pseudo ground-truth. When the teacher is wrong about the *class* of a well-localised box, the wrong label enters CL and the model actively learns to mis-classify.

The MTL setting provides an independent second opinion on the class: the segmentation head’s own posterior over the box interior. CTPV consults it. For each pseudo-box b of class c :

$$q(b, c) = \frac{1}{|\Omega_b|} \sum_{i \in \Omega_b} p_{i, \sigma(c)}^{T, \text{seg}}$$

is compared to a threshold $\tau_{\text{CTPV}} = 0.3$. Boxes with $q < \tau_{\text{CTPV}}$ are removed from the pseudo-label pool *before* CT-CL, CT-CR, or the detection loss are computed on this step.

CTPV is a **pseudo-label-level** XVA primitive: it acts before any loss, filtering the supervision signal itself by cross-task agreement. It has no analog in Moraiti *et al.* 2026 for the same reason CT-CR does not — there is no second head to verify against. On ACDC continual (see §9) CTPV alone contributes **+0.23 AP50 / +0.28 mIoU** over the CT-CL + CT-CR + STR baseline, with the largest gains on snow (+0.65 AP50 / +0.84 mIoU) where teacher-drift accumulates most.

7. Fault-mode framing — a diagnostic reformulation of the prior narrative

The prior paper motivated its three components (MT, CL, SR) as complementary techniques. Our extension re-frames the discussion around two **structural failure modes** of teacher-student CTTA:

- **FM-1 (Teacher-collapse under a single supervision signal).** In detection, teacher scores fall below the confidence floor on hard weather and no pseudo-boxes survive; adaptation silently stops. In segmentation, teacher per-pixel entropy grows and the soft-CE loss amplifies teacher noise into gradient noise. Both are visible in single-head systems because the only supervision channel is the head that is drifting.
- **FM-2 (Drift under uniform restore).** Uniform-Bernoulli SR treats shared trunk and task heads identically. The shared trunk, on every gradient path, accumulates domain-specific bias faster than the mask reverts it.

CT-CL, CT-CR, and CTPV target FM-1 at three levels of granularity (instance, dense, pseudo-label). STR targets FM-2. This framing is a *sharper* version of the observation in Moraiti *et al.* 2026 that “MT alone suffers from noisy pseudo-labels” and “adaptation for a long time leads to catastrophic forgetting”. The prior paper diagnosed the symptoms; the extension diagnoses the mechanisms and shows that both are structural consequences of single-head CTTA.

8. Framework specialisations

The MTL meta-architecture supports three deployment modes controlled by config flags. Each specialisation is a strict subset of the full XVA framework:

Variant	Source	Active components	Relationship to Moraiti <i>et al.</i> 2026
CTCMT_Det	Mask R-CNN R-50 FPN	MT + soft-CE det consistency + STR	direct port of the prior single-task recipe to a two-stage det backbone; CT-CL falls back to object-level SupCon

Variant	Source	Active components	Relationship to Moraiti <i>et al.</i> 2026
CTCMT_Seg	Semantic FPN R-50	MT + soft-CE seg + STR + per-class prototype anchor + trimmed 6-view aug-avg	prior recipe transposed to segmentation (a scenario not tested in the prior paper)
CTCMT-MTL + V2 + CTPV	Panoptic-FPN R-50 MTL	soft-CE det + soft-CE seg + CT-CL + CT-CR + STR + CTPV	the paper’s main contribution

The single-task variants exist to (a) verify that the framework does not regress on prior workloads and (b) provide fair same-source-model baselines for the MTL headline. In practice CTCMT_Det on Mask R-CNN source beats AMROD by +0.92 AP50 on ACDC continual, and CTCMT_Seg on Semantic FPN source beats CoTTA by +1.28 mIoU — both consistent with the prior paper’s finding that MT + CL + SR outperforms plain MT.

9. Empirical validation on ACDC continual

We report a fresh ablation on ACDC continual (fog → night → rain → snow), 3 seeds per row, using the Panoptic-FPN R-50 MTL source. **Baseline** = MTL soft-CE det + soft-CE seg + uniform stochastic restore (an MTL port of the plumbing that would result from applying the prior work’s components naively to a shared-backbone model).

Increment	AP50	mIoU
MTL baseline (uniform SR, no CT-CL, no CT-CR, no CTPV)	34.09 ± 0.05	31.79 ± 0.06
+ STR (task-aware SR, replaces uniform SR)	34.89 ± 0.07	32.41 ± 0.07
+ CT-CL (three-way SupCon over det + seg views)	35.98 ± 0.12	33.23 ± 0.05
+ CT-CL + CT-CR + STR (three-component XVA)	37.20 ± 0.06	34.14 ± 0.03
+ CT-CL + CT-CR + STR + CTPV (full)	37.43 ± 0.07	34.42 ± 0.06
Full delta over baseline	+3.34	+2.63

Every increment is significant against seed variance ($\gg 3\sigma$). The STR row alone reproduces the “SR is the dominant single component” finding from the prior paper — even on this new backbone and this new benchmark — but the addition of CT-CL, CT-CR, and CTPV together adds another **+2.54 AP50 / +2.01 mIoU** on top, confirming that the MTL-specific mechanisms deliver gains that were impossible to access from the single-task formulation.

Foggy Cityscapes (single-shot). Same source model: - Source only: 32.15 AP50 / 44.12 mIoU - Full extension: **33.19 AP50 / 49.97 mIoU**

Comparison against SOTA on ACDC continual. Full-extension vs published methods on the same Mask R-CNN R-50 source (Table 1 of the paper): W3TTA-OD (Yoo *et al.* CVPR 2024) 34.20 AP50; AMROD (Wei *et al.* ESWA 2026) 35.34 AP50; ours **36.06** (single-task

CTCMT_Det+STR) and **37.43** (full MTL XVA). Full-extension vs CoTTA on ACDC continual seg (same Semantic FPN source): CoTTA 36.39 mIoU; CTCMT_Seg **37.67 mIoU**.

10. What is new vs. what is preserved

Component in this paper	Origin	Relationship to Moraiti <i>et al.</i> 2026
Mean Teacher skeleton EMA teacher update, $\alpha=0.9998$	Tarvainen & Valpola 2017 Moraiti <i>et al.</i> 2026	preserved verbatim preserved (same rate)
Dynamic per-class thresholds	AMROD (Wei 2026), also used in prior CT-CMT	preserved verbatim
Score-EM confidence gate	AMROD (Wei 2026)	preserved verbatim
Object-level SupCon on det features	Moraiti <i>et al.</i> 2026 §3.3	preserved as the det $\leftrightarrow$ det pair type inside CT-CL
Cross-task views (det + seg per bbox)	this paper §3	new: seg view is a novel construction
CT-CL (three-way SupCon: det$\leftrightarrow$det, seg$\leftrightarrow$seg, det$\leftrightarrow$seg)	this paper §3	new: reduces to prior CL when seg branch is absent
CT-CR (dense bbox $\rightarrow$ seg-class CE)	this paper §4	new: no analog in prior work
Uniform Bernoulli stochastic restore	Moraiti <i>et al.</i> 2026 §3.2 (from CoTTA)	preserved as the $\eta = 1$ special case of STR
STR (shared-trunk vs task-head split)	this paper §5	new generalisation: recovers prior uniform SR when the shared/task distinction is trivial
CTPV (seg-verifies-det pseudo-label filter)	this paper §6	new: no analog in prior work
Backbone choice (Panoptic-FPN R-50 MTL)	Kirillov <i>et al.</i> 2019	new source model (prior work: YOLOX single-task)
Benchmark (ACDC continual + Foggy CS MTL)	Sakaridis <i>et al.</i> 2021	new evaluation setting; prior work used SHIFT / KITTI / Cityscapes / CLAD-D / COCO-C

The extension therefore preserves the entire prior framework as a substructure, generalises its two flagship mechanisms (CL and SR) into constructs that reduce to the originals in the single-task limit, and introduces two mechanisms (CT-CR, CTPV) that are only definable when a second head is available. The MT skeleton, dyn-threshold, and score-EM gate that the prior work adopted from other CTTA methods are inherited without change.

11. Practical remarks

1. **The prior work’s implementation stays valid.** The CT-CMT adapter in [shift-tta/shift-detection-tta](#) uses YOLOX PAFPN features for RoIAlign, and the object-level CL is applied over teacher-pseudo boxes. Replacing YOLOX with Panoptic-FPN and adding a segmentation branch is a strict feature addition — no part of the prior implementation needs to be removed.

2. **Hyperparameters carry over.** SupCon temperature $\tau = 0.07$, EMA decay $\alpha = 0.9998$, dyn-threshold τ_c init, min, max are used unchanged from the prior paper. New hyperparameters ($\eta = 0.1$ for STR, $\lambda_{CL} = 0.5$, $\lambda_{CR} = 0.3$, $\tau_{CTPV} = 0.3$) were tuned on a single held-out run and fixed across all reported experiments.
3. **The paper cites Moraiti *et al.* 2026 as its immediate predecessor** and positions itself as the MTL extension of that framework.

12. Section-level correspondence with Moraiti *et al.* 2026

Prior section	Prior content	This paper’s corresponding section
§1 Introduction	Motivation for continual TTA; catastrophic forgetting	§1 Motivation and continuity
§2 Related work	UDA, TTA, continual TTA	Merged into §7 (fault-mode framing) and §10 (component-attribution)
§3.1 MT framework	Teacher/student EMA, consistency loss	§1.1 “Teacher-student plumbing” — preserved verbatim
§3.2 Stochastic Restoration	Uniform Bernoulli mask, $p = 0.01$	§5 STR — generalised (backward-compatible)
§3.3 Object-level CL	RoIAlign features, class-based SupCon	§3 CT-CL — generalised (backward-compatible)
§3.3.3 Multi-scale feature maps	Three PAFPN scales summed	Retained; FPN scales P3/P4/P5 in Panoptic-FPN
§4 Experiments	SHIFT, KITTI, Cityscapes, CLAD-D, COCO-C	§9 — evaluation on ACDC continual + Foggy CS
§5 Conclusion	Positive contributions of MT + CL + SR	Restated and sharpened in §7 (FM-1 / FM-2 diagnostic)
— (new)	—	§4 CT-CR (dense cross-task consistency) — new mechanism
— (new)	—	§6 CTPV (pseudo-label verification) — new mechanism

The extension is a linear supersession of the prior paper’s methodology: prior components appear as either preserved plumbing or as backward-compatible generalisations. Two new mechanisms (CT-CR, CTPV) close failure modes that are only diagnosable in the MTL setting.